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A Computer Assisted Composition system for a live coding environment

Poster

I am developing a Computer Assisted Composition (CAC) system for use in the Impromptu interactive programming environment, developed by Andrew Sorenson. The Impromptu interface utilises the Scheme programming language, offering the composer/performer a flexible starting point to construct almost anything the Lisp programming language will allow. I am developing a library of algorithmic tools at a 'higher level' of abstraction with a view to assist in the generation of musical structures.

Live coding is the performance practice of creating artistic work by writing and modifying the computer programs that generate the work during the performance. There are emerging aesthetics in the field of live coding which I will need to consider in relation to the development of my CAC system. For example, there is a preference amongst live coders to allow code to be clear enough such that underlying algorithms can be deciphered.

There is also some debate concerning how 'high level' such a system should be. At one end of the 'high level' spectrum, packages with pre-scripted interfaces are generally eschewed by live coders largely because they don't offer access to underlying algorithms, and/or are not transparent in their presentation of those algorithms. At the 'low-level' end, building 'from scratch' with original code can be daunting, and not always advantageous in a performance situation. These are some of the issues I am contemplating in the design of my CAC system.

Any system of compositional tools will ultimately reflect the compositional ideology of the developer. Given this, rather than attempt to create a 'musically neutral' system intended to accommodate existing compositional practices, I am pursuing a compositional strategy which my system is aimed to reflect. My compositional strategy is still being formulated, however one of my concerns is to provide access and control to both micro and macro musical structures. While there are many exceptions, in general, the distinction between micro and macro structures has divided the approach to composition of many CAC's. By offering consistent methods to control musical structure at any level, my system aims to blur the distinction between micro and macro structures.

I am interested in defining the relationship between structures. This has evolved from an ongoing interest in linguistics and 'concrete poetry'. Of particular interest are poetic techniques that are based upon - sometimes ambiguous - relationships between texts. In such cases it is the relationship itself that becomes the basis of the work, as opposed to a specific meaning denoted by the text itself. Employing techniques such as these offers useful possibilities for a compositional approach to musical structures.

In a live coding context my CAC system would offer the composer/performer the potential to construct and organise material on a number of levels at once. While the compositional system employed reflects the interests of the developer, it is hoped that the tools provided are flexible enough, and of interest to other composers who want to realise their own compositional approach in a live performance context.